Low Temperature SoIC Bonding and Stacking Technology for 12-/16-Hi High Bandwidth Memory (HBM)

M. F. Chen, C. H. Tsai, Terry Ku, W. C. Chiou, C. T. Wang, and Douglas Yu, *Fellow, IEEE*

***Abstract*— A low temperature system-on-integrated-chip (SoIC) bonding and stacking technology is proposed and implemented for 3-D memory integration, such as 3-D static random access memory (SRAM) or dynamic random access memory (DRAM) cube. It extends not only 2-D pin count to 12 000/mm^2 but also overcomes the obstacle of the stacking height. 4-Hi, 8-Hi, and 12-Hi stacks, each with 1 base die and 4, 8, and 12 dies, respectively, are realized and demonstrated in this article. The daisy chains in the 4-Hi/8-Hi/12-Hi structures incorporating over 10 000 through silicon vias (TSVs) and bonds are tested with liner $I-V$ curves, which indicates the good bonding and stacking quality. The electrical link from base controller to top DRAM for 12-Hi and 16-Hi high bandwidth memory (HBM) structure is built up to study the bandwidth (BW) and power efficiency. Compared to μbump technology, the BW for the 12-Hi and 16-Hi structures using SoIC bonding shows the improvement of 18% and 20%, respectively, under same pin pitch and the power efficiency has the improvement of 8% and 15%. For thermal performance, the 12-Hi and 16-Hi SoIC-bond structure is 7% and 8% better than those using μbump technology, respectively. With this innovative SoIC bonding and stacking technology, the bond pitch is scalable to sub-micrometer and the die thickness is manageable to be thinner, which are prospected for the application of higher BW 3-D memory with tera byte (TB)/s per stack.**

***Index Terms*—3DIC, bonding, dynamic random access memory (DRAM), high bandwidth memory (HBM), Moore's law, static random access memory (SRAM), system-on-integrated-chip (SoIC), system scaling, wafer level system integration (WLSI).**

I. Introduction

MOORE'S law evolves from transistor scaling alone onto system-level scaling [1], while it becomes inevitable to integrate multiple functions/blocks/chips in enormous system with sustainable performance gain, power density, and cost benefits. To achieve the sustainable system performance/power with a competitive cost in addition to the reintegration of chiplets partitioned from large SoC [2], algorithm-specific accelerators associated with 3-D memory are proposed next to the CPU. Recent flourish of artificial intelligence (AI) accelerator or GPU systems are examples. They strongly require high bandwidth (BW) and high capacity 3-D memory to support massively parallel processing and efficiently convolution computing. Among these systems, 3-D memory plays an indispensable role to narrow down the performance gap between the required memory BW and the practical memory BW [3].

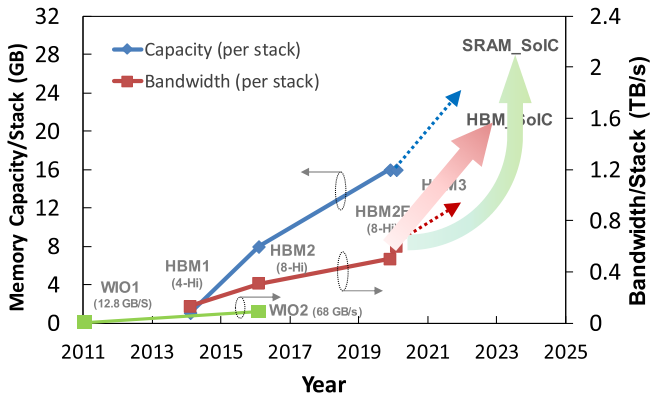

Fig. 1. 3-D memory system trend, including wide IO and HBM.

Fig. 1 shows the evolutions of 3-D memory, including wide I/O and high bandwidth memory (HBM). The concept of 3-D memory was first demonstrated and presented in 2011, so-called wide I/O [4], [5]. It stacked and packaged multiple dynamic random access memory (DRAM) cores on the main computing logic core through silicon via (TSV), μbump, and underfill technologies. Although it could be theoretically stacked with more tiers and formed into a low profile 3-D computing system, it would suffer thermal problem since main heat source from the logic core could not effectively spread out, which inevitably damaged the retention performance of thermal sensitive DRAM cores. In 2014, HBM was proposed [6], [7], and adopted to integrate with computing logic cores side-by-side on the interposer, as shown in Fig. 2 [8]. Similar to wide I/O, it is realized by TSV formation, μbump joint, and underfill technologies. It incorporated one base logic stacked with multiple DRAM cores. Different from wide I/O, the computing power of its base logic, contented with testing circuits, buffers, and I/O PHY, was eliminated. Currently, it can provide stacking height of DRAM cores up to eight and the BW up to 500–600 GB/s, respectively [9], [10].

Despite these, the higher performance HBM is still strongly demanded. Current stacking technology with μbump joint

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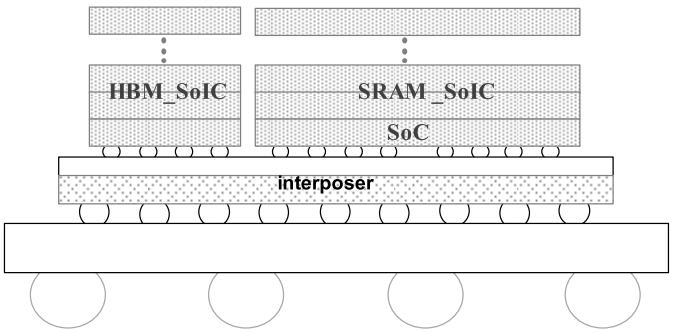

Fig. 2. System integration of 3-D memory: HBM integrated with static random access memory (SRAM)/SoC cubes.

limits the development of higher BW and capacity HBM from more sacking number and higher power efficiency. The BW related to I/O pin counts and the capacity related to stacking number had been limited by the μbump structure. Furthermore, the μbumps with underfill introduce parasitic capacitance, resistance, and inductance, as well as thermal resistance, degrading the data rate per pin and thermal performance. These above make it difficult to form higher BW over tera byte (TB)/s per stack and higher stacking number, like 12-Hi or above. Consequently, a new bonding and stacking technology is needed to break through the barriers from μbump.

In this article, a low temperature bonding and stacking technology is proposed to realize multi-Hi structures. It leverages system-on-integrated-chips (SoICs) process and adopts foundry front-end wafer technology [11], [12]. The 4-Hi/8-Hi/12-Hi SoIC-bond structures are manufactured and the corresponding daisy chains with over 10 000 TSVs and SoIC-bonds are verified. Furthermore, 12-/16-Hi HBM is taken for case studies. The electrical link and thermal path of the 12- and 16-Hi SoIC-bond HBM structures from a base logic die to top DRAM core are modeled to demonstrate their benefit on the BW, energy per bit, and junction temperature against those using μbump technologies. The scalability of the proposed bonding and stacking technology is projected for the application of 3-D cache memory and higher BW HBM. Finally, the reliability performance is demonstrated.

II. SoIC Technology and Structure

A. Standard SoIC Bonding

SoIC, a 3-D integration technology, is proposed for the integration of chiplets and functional dies. It implements SoC partition and reintegrates chiplets into new SoC system. It leverages advanced foundry wafer front-end processes and applies thin die handling and bonding processes. As shown in Fig. 3, it can be a face-to-face or face-to-back stacking. The adoption of face-to-face or face-to-back stacking is determined by the trade-off of signal integrity (SI)/power integrity (PI)/thermal performance and cost.

Fig. 4 shows the comparison of integration flow for SoIC and typical 3DIC. SoIC utilizes SoIC bond to realize bumpless and underfill-less interconnection between two dies, while typical 3DIC uses μbump process with solder joint and underfill to form a die stacking structure. Against typical 3DIC,

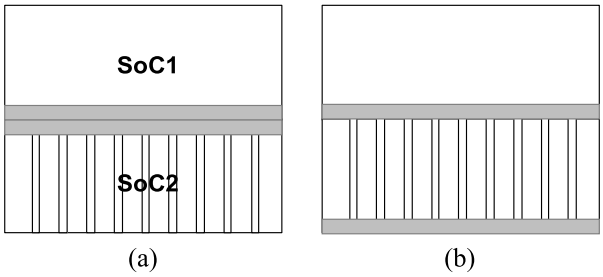

Fig. 3. (a) Face-to-face and (b) face-to-back SoIC bonding process.

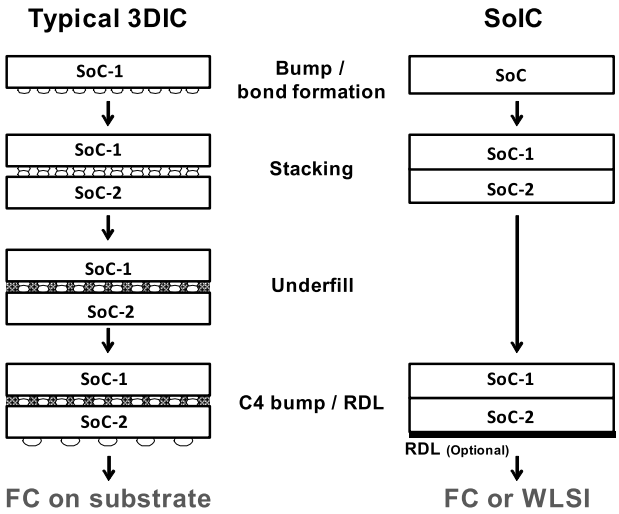

Fig. 4. Comparison of integration flow for typical 3DIC and SoIC.

SoIC achieves high integration yield and reliability from the adoption of less material and process steps.

B. Low Temperature SoIC Bonding

In contrast with the standard SoIC bonding, a low temperature SoIC bonding technology was developed for the integration of thermal sensitive dies like DRAM cores. The bonding temperature is less than 300 °C and it is comparable to the reflow process of the lead-free solder joint. A razor blade test is conducted to measure the bonding strength. The bonding strength is more than 2.5 J/m^2, which is close to the bonding strength using high temperature bonding process. Furthermore, to verify the bonding quality, the daisy chains of the bonding structures corresponding to the standard and low temperature bonding processes are tested. The measurement result depicts that the low temperature bonding process has the chain resistance similar to that for standard high temperature bonding process in Fig. 5. This developed low temperature bonding process demonstrates similar mechanical and electrical performance to the high temperature bonding process, which can mitigate the thermal budget issue from standard high temperature process and also show the merit of cost reduction.

C. 3-D Stacking

To achieve 3-D memory system, multiple die stacking using low temperature bonding and stacking is proposed. For multiple die stacking, face-to-back stacking was chosen for the optimized design and cost, as shown in Fig. 6(a). To achieve the multi-Hi SoIC-bond structure, two stacking flows, TSV

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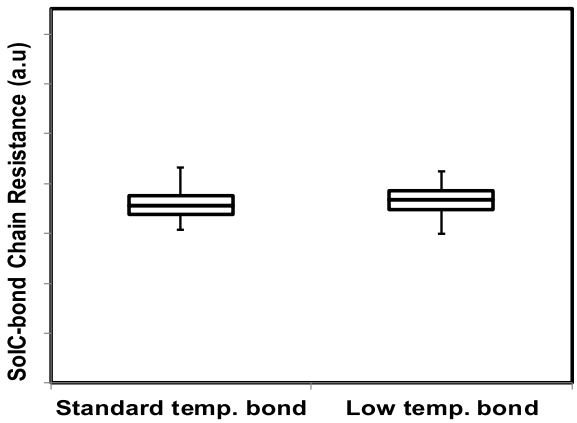

Fig. 5. Measured SoIC-bond daisy chain resistance for standard temperature bonding process and low temperature bonding process.

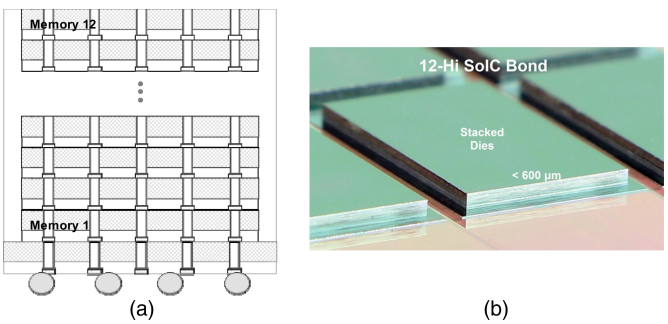

Fig. 6. (a) 12-Hi face-to-back die stack using SoIC bond. Base die is larger than other stacked dies. Stacked die are interconnected through TSVs, BEOL, and SoIC bonds. (b) Photograph of 12-Hi SoIC bond structure with the total height below 600-μm. The height can be adjusted to fit the system requirement.

reveal last and TSV reveal first, had been developed. The flow for TSV reveal last process allows the die with a modest thickness to be bonded first before it is thinned down to reveal the TSV. On the contrary, the flow for TSV reveal first process starts from wafer thinned down to reveal TSV and then diced to discrete dies, followed by bonding. The former one is simple, but suffers die corner/edge rounding problem, while the latter one is free from rounding problem, but faces surface contamination problem. Both of the two stacking flows have been optimized to offer more flexibility in the multiple die stacking and integration. By the low temperature SoIC bonding and face-to-back stacking, a 12-Hi SoIC-bond structure, which is composed of 1 base die and 12 stacked dies, is realized with total height smaller than 600-μm, as shown in Fig. 6(b). This indicates the feasibility of 3-D stacking.

III. Performance and Scalability

The low temperature SoIC bonding and stacking was successfully developed. In the following, the bonding quality in terms of chain resistance, system performance of bandwidth and energy per bit, and thermal performance will be demonstrated. Moreover, the scalability including bond pitch, stacking number, and die size will be investigated to extend the 3-D memory application.

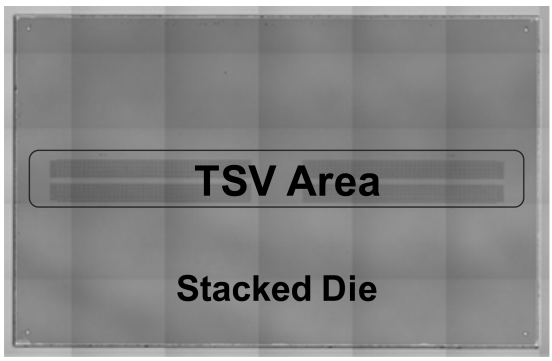

Fig. 7. Optical microscope (OM) images of one of the stacked dies in the multi-Hi structure.

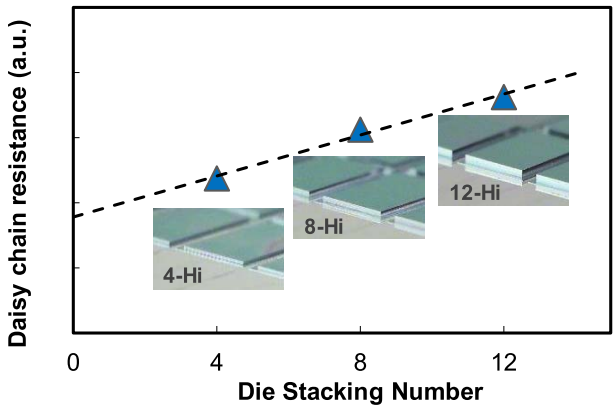

Fig. 8. Daisy chain resistance comparison for 4-Hi, 8-Hi, and 12-Hi die stacking. The daisy chain incorporates TSV, BEOL, and SoIC bond.

A. DC Performance

To check integration integrity of the low temperature SoIC bonding and stacking, the interconnect chains incorporating TSV, back end of line (BEOL), and bond are fabricated with bond pitches of 36 and 9-μm in different stacking height samples. Different pitch could be needed for different 3-D memory architecture design. 4-Hi, 8-Hi, and 12-Hi structures are implemented with 1 base die and 4, 8, and 12 smaller dies, respectively. The bond pitch of these structures is 36-μm. Each smaller die mimics an HBM architecture with TSVs in the center of die as shown in Fig. 7. The $I - V$ curves are measured and the chain resistance is obtained. The corresponding interconnect chains are tested with a low dc resistance for each structure and demonstrated with a linear resistance behavior between 4-Hi, 8-Hi, and 12-Hi structures, as depicted in Fig. 8. Also, three types of daisy chains from the bottom die to top die in 12-Hi structure are also tested. Fig. 9 shows the testing results. The long chains composed of over 10 000 TSVs, BEOL, and bonds demonstrate linear $I - V$ curves with a reasonable bonding resistance, which indicates interconnect chains are continuous links with excellent bonding and stacking quality. For bond pitch of 9-μm, a 4-Hi die stacking structure with four same dies is realized. It can be considered as a SRAM cube. A steep accumulation line for chain resistance is observed in Fig. 10, which indicates the chain resistance across different 4-Hi structures is almost the same. These

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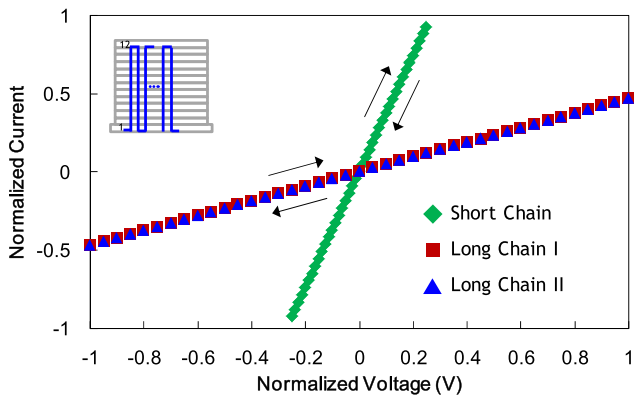

Fig. 9. Measured $I-V$ curves for 12-Hi SoIC-bond structure with TSV, BEOL, and bonding structures.

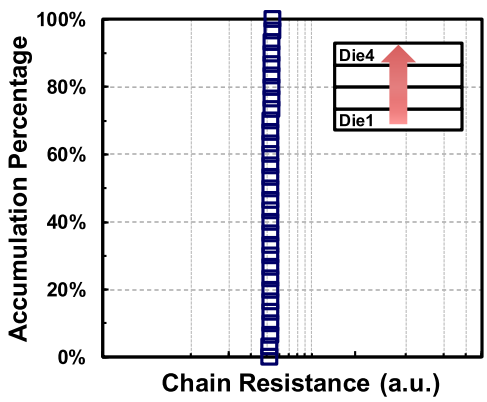

Fig. 10. Chain resistance comparison for 4-Hi die stacking. The chain incorporates TSV, BEOL, and SoIC bond.

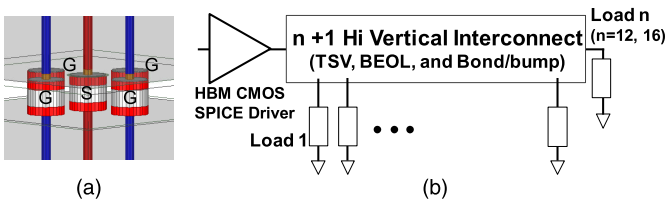

Fig. 11. (a) 3-D model of μbump structures in HFSS and (b) electrical model from a base logic die to top DRAM core.

above demonstrate the process stability and feasibility of the proposed low temperature bonding and stacking technology.

B. Electrical and System Performance

To characterize the electrical and system performance of the proposed stacking structures, the signal transmission path from base logic to top memory core is modeled as shown in Fig. 11. 12-/16-Hi SoIC and μbump structures with the pitch of 36-μm are simulated in Ansys high frequency structure simulator (HFSS), a 3-D electromagnetic tool. Fig. 11(a) shows the modeled μbump structure comprising one signal path and four shielding ground paths. Fig. 11(b) describes the electrical link, which is built up in HSPICE to study the BW and power efficiency.

Due to lower parasitic effect of SoIC bond as depicted in Table I and lower insertion loss of the electrical link as shown

TABLE I
PARASITIC R/L/C PARAMETERS

Symbol	μbump	SoIC-bond
R (mΩ)	1x	0.2x
L (pH)	1x	0.15x
C (fF)	1x	0.12x

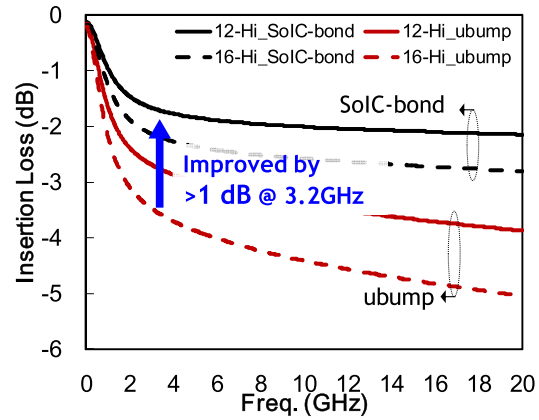

Fig. 12. Insertion loss for the interconnects of 12-Hi/16-Hi μbump structures and SoIC-bond structures.

in Fig. 12, the proposed 12-Hi structure demonstrates the improvement of 18% and 8% on the BW and energy per bit, respectively. For 16-Hi SoIC-bond structure, the improvement of 20% and 15% on BW and energy per bit is observed, respectively. Both results are shown in Fig. 13. The estimation of BW is to read eye diagram at the last load and the maximum data rate under an acceptable eye open while power efficiency is evaluated based on the power consumption at the same data rate for both structures. Note that CMOS SPICE driver is put at the base die and multidrop architecture with a capacitive loading at each die is assumed [7].

C. Thermal Performance

Fig. 14(a) depicts the diagram of thermal modeling environment with one processor and four SoIC-bond structure on the interposer. It was conducted by Icepak, a commercial computational fluid dynamics. For thermal performance comparison, the local model of local interconnect, including μbump and SoIC bond, was built up to extract the effective thermal conductivities of the bonding layers and equip the effective thermal conductivities of the layers corresponding to their thickness in the global model. A heat transfer coefficient 10 000 W/C-mm^2, mimicking liquid cooling, is applied on the top surface of the package while the other surfaces are summed to be adiabatic. Since the thermal barrier of μbump is eliminated, the 12-Hi and 16-Hi SoIC-bond structures show 7% and 8% lower in ΔT than 12-Hi and 16-Hi μbump structures. ΔT is defined by the difference between base die junction, T_j, and ambient temperature, T_a, T_j–T_a. The lower the temperature difference is, the better thermal performance

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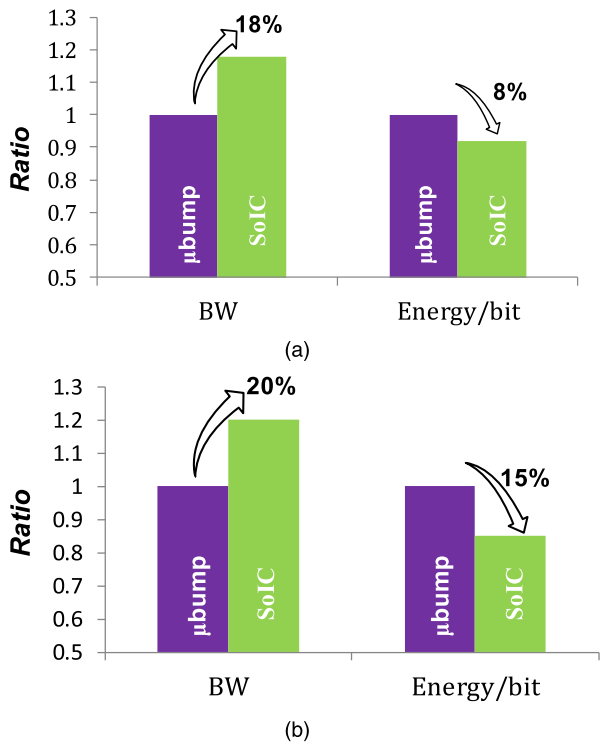

Fig. 13. BW and energy-per-bit improvement for (a) 12-Hi and for (b) 16-Hi when compared to 12-Hi and 16-Hi μbump structures.

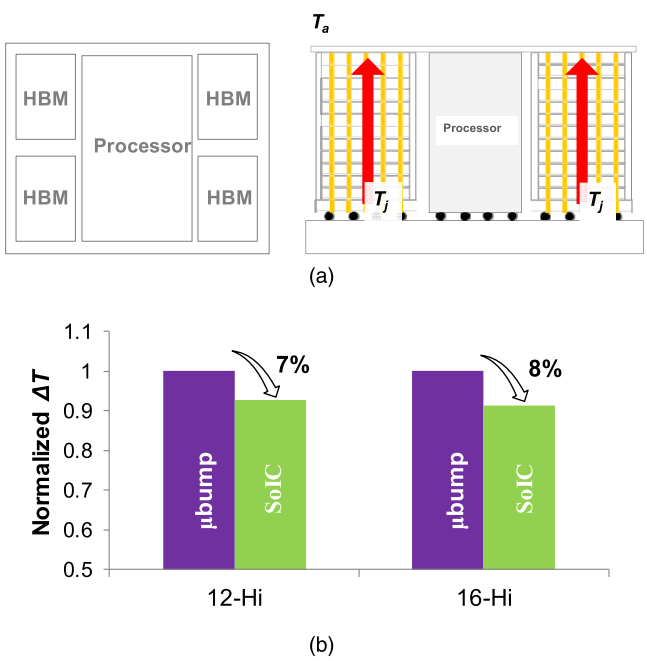

Fig. 14. (a) Thermal modeling environment with one processor and four HBM cubes and (b) ΔT comparison. ΔT is defined by the difference between base die junction temperature, T_j and ambient temperature, T_a.

is. With the bond density refinement and die thickness thinned down, thermal performance of SoIC can be further enhanced.

D. Scalability

To further extend the BW of HBM, the proposed technology can scale down not only bond pitch to accommodate more pin counts but also die thickness to reduce the TSV parasitic capacitance, which benefits the BW in terms of the product of data rate and pin counts. As shown in Fig. 15, SoIC bond can have the pitch scale up to sub-10-μm and even micrometers without the need of solder joint, whereas for conventional flip-chip C4 bump, the bump pitch may be over 100-μm and for 3DIC technology, the μbump pitch may be 40-μm or around. If sub-10-μm SoIC bond pitch is applied, 16 times denser pin density than that using current μbump technology can be achieved. It becomes easy to achieve double BW even if the driver speed is not enhanced.

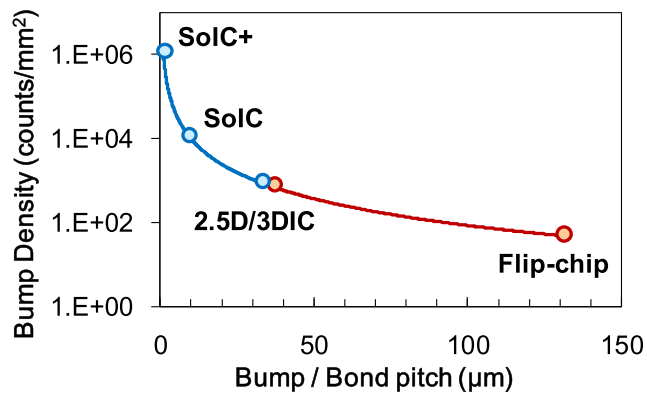

Fig. 15. I/O density versus bump/bond pitch for flip-chip, 2.5DIC, 3DIC, and SoIC.

Also, each die thickness in the proposed bonding and stacking technology can be scaled to be thinner and help to enhance the data rate per pin and reduce the energy per bit, as shown in Fig. 16. Moreover, the die thickness of scalability may determine the capacity related to stacking number HBM stack can have. If the die thickness is reduced, it can help to increase stacking number under the same Z-height. In short, the scalability of die thickness from SoIC bond structure can benefit the BW and capacity of high BW 3-D memory.

IV. Reliability Performance

In contrast with μbump joints applied for the multi-Hi stacking structure, the low temperature SoIC bonding and stacking technology provides a bumpless solution without using epoxy material, which can directly mitigate the reliability concerns in the μbump joints for 3-D memory structure. The breakdown voltage and thermal cycling test are executed for reliability verification. The breakdown voltages (V_{bd}) for the 12-Hi SoIC-bond structure, as shown in Fig. 6(b), were measured. Fig. 17 shows the samples for the 12-Hi have similar breakdown voltage, which demonstrates the structural integrity of the interconnect chains. For thermal cycling test, the interconnect chain resistance in the 12-Hi samples is measured at time zero (T_0) and after 200 cycles under Thermal-Cycling Test B (TC-B) of JEDEC JESD22-A104, respectively. Fig. 18 shows the data at TC-B 200 cycles are consistent with the data at T_0, which indicate the good reliability of the interconnect chain using the proposed low temperature SoIC technology.

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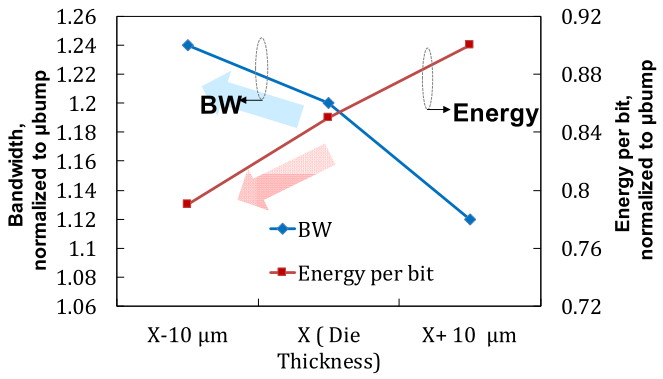

Fig. 16. BW and energy/bit evaluation for 16-Hi SoIC-bond structures with different die thickness. X is the thickness used for electrical and thermal modeling of 16-Hi.

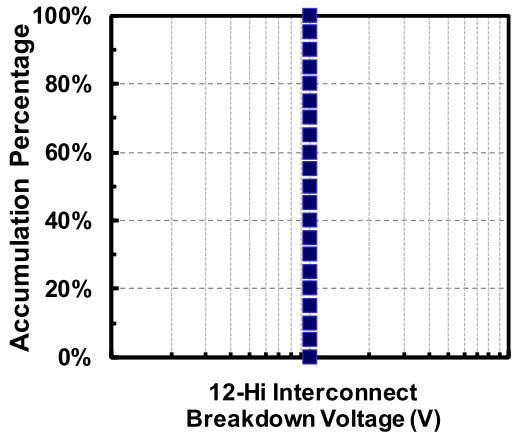

Fig. 17. Breakdown voltage of interconnect for 12-Hi die stacking, where consistent V_{bd} is presented across different samples.

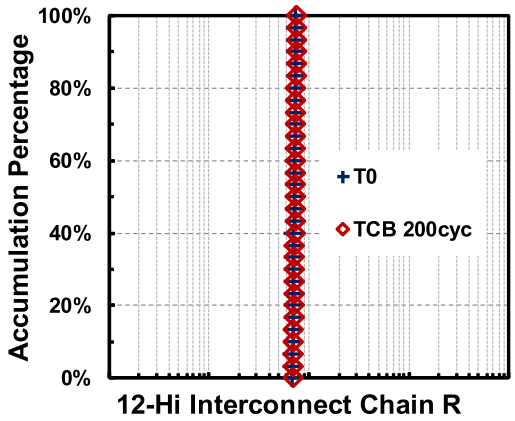

Fig. 18. Comparison of interconnect chain resistance under T_0 and TCB-200 for the 12-Hi samples using low temperature bonding and stacking technology.

V. Conclusion

Four-Hi/8-Hi/12-Hi structures using low temperature SoIC bonding and stacking technology are realized for bond pitch of 36 and 9-μm and continuous interconnect chains incorporating over 10 000 TSVs and bonds are proved. The electrical and thermal performance shows 18%, 8%, and 7% improvement on BW, power efficiency, and ΔT for 12-Hi structure, respectively. Most importantly, the scalability in bond pitch up to sub-micrometer and stacking number with over 12-Hi could be prospected for future 3-D memory system with TB/s BW per stack.

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